

Marker	Measures	Ideal Range	Date	Date	Date	
TG/HDL ratio	Insulin resistance	< 1				Test Result
AA/EPA ratio	Balance of inflammation to resolution	1.5-3				Test Result
HbA1c	Activation of AMPK	4.9-5.1%				Test Result
HOMA-IR	Estimates insulin resistance, a predictor of future chronic disease, and surrogate marker for AMPK	< 1				Test Result

1. *Triglyceride/HDL cholesterol (TG/HDL) ratio*: This is a surrogate marker of insulin resistance. This marker can be significantly reduced by following a balanced diet.
2. *Arachidonic acid (AA) to eicosapentaenoic acid (EPA) ratio*: This is a marker of the balance of unresolved inflammation. This marker is primarily reduced by high-dose omega-3 fatty acids.
3. *Glycosylated hemoglobin (HbA1c)*: This is marker of long-term blood glucose control which will indicate the degree of inhibition of AMPK as elevated blood glucose levels inhibit AMPK activity. This marker is primarily reduced by high-dose polyphenols.
4. *HOMA-IR*: Insulin resistance is assessed by calculating HOMA-IR using the formula **fasting insulin (microU/L) x fasting glucose (nmol/L)/22.5** and there are many calculators on-line.

EPA/DHA suggested dosage

No chronic disease	2.5g
Overweight/obese, type II diabetes, CHD etc	5g
Chronic pain	7.5g
Neurological disorders	10g

Polyphenol suggested dosage

- 500mg helps reduce oxidative stress
- 1000mg helps reduce inflammation
- 1500mg helps reduce the rate of aging and increases mitochondrial synthesis